A COVERING STRUCTURE ON AN ABELIAN CATEGORY MAKES IT A SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete verification of the site axioms; only the base-change step is attributed to a named result carried on this card as a navigation stub.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Covers of surjective morphisms. Same chapter, the declaration of which families of morphisms are coverings in the construction. Supplies the coverings whose axioms are verified.
- Base change of surjections. Homology, Lemma homology-lemma-epimorphism-universal-abelian-category. Supplies that the base change of a surjective morphism is surjective, which is the axiom verified by citation.

Source reference. Original definition 00VH: sources/stacks-sites/pinned-original-source.tex, source lines 896–896. Navigation pointer invoked at proof line 1274; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05PK: sources/stacks-homology/pinned-original-source.tex, source lines 1136–1136. Navigation pointer invoked at proof line 1283; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:injectives.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let*

$$Cov = \{\{f : V \to U\} \mid f \text{ is surjective}\}.$$

Then $(\mathcal{A}, Cov)$ is a site, see Sites, Definition 00VH.

Proof. Note that $\mathrm{Ob}(\mathcal{A})$ is a set by our conventions about categories. An isomorphism is a surjective morphism. The composition of surjective morphisms is surjective. And the base change of a surjective morphism in $\mathcal{A}$ is surjective, see Homology, Lemma 05PK. □